

Succinylcholine (suxamethonium): Drug information - UpToDate

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For additional information see "Succinylcholine (suxamethonium): Patient drug information" and "Succinylcholine (suxamethonium): Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

ALERT: US Boxed Warning

Risk of ventricular dysrhythmias, cardiac arrest, and death from hyperkalemic rhabdomyolysis in pediatric patients:

Acute rhabdomyolysis with hyperkalemia followed by ventricular dysrhythmias, cardiac arrest, and death has occurred after the administration of succinylcholine to apparently healthy pediatric patients who were subsequently found to have undiagnosed skeletal muscle myopathy, most frequently Duchenne muscular dystrophy.

When a healthy appearing pediatric patient develops cardiac arrest within minutes after administration of succinylcholine not felt to be due to inadequate ventilation, oxygenation, or anesthetic overdose, immediate treatment for hyperkalemia should be instituted. In the presence of signs of malignant hyperthermia, appropriate treatment should be instituted concurrently.

Reserve the use of succinylcholine in pediatric patients for emergency intubation or instances where immediate securing of the airway is necessary (eg, laryngospasm, difficult airway, full stomach) or for IM use when a suitable vein is inaccessible.

Brand Names: US

- Anectine;
- Quelicin

Brand Names: Canada

- Quelicin Chloride

Pharmacologic Category

- Neuromuscular Blocker Agent, Depolarizing

Dosing: Adult

Note: Dose to effect; doses will vary due to interpatient variability. Neuromuscular blockade does not provide pain control, sedation, or amnestic effects. Appropriate analgesic and sedative medications should be used before and during administration of neuromuscular blockade to achieve deep sedation. To avoid prolonged neuromuscular blockade, use carefully and/or consider dose reduction in patients with reduced plasma cholinesterase activity due to genetic abnormalities of plasma cholinesterase or associated with other conditions (eg, electrolyte abnormalities, neuromuscular disease, severe liver or kidney disease, malignant tumors).

Electroconvulsive therapy, muscle relaxation

Electroconvulsive therapy (ECT), muscle relaxation (off-label use): IV: 0.5 to 1.5 mg/kg (doses ≥ 0.75 mg/kg are more common in clinical practice); adjust dose higher or lower based on muscle mass, muscle relaxation with previous ECT, and patient-specific risks associated with sub-optimal relaxation (eg, increase dose in patients with osteoporosis) (Ref). **Note:** Most patients will recover respiratory function and muscle strength within 9 to 10 minutes (Ref).

Neuromuscular blockade for endotracheal intubation, surgery, or mechanical ventilation

Neuromuscular blockade for endotracheal intubation, surgery, or mechanical ventilation (as adjunct to general anesthesia):

IM: Up to 3 to 4 mg/kg, maximum total dose: 150 mg.

IV:

Intubation: 0.6 mg/kg (range: 0.3 to 1.1 mg/kg).

Rapid-sequence intubation (off-label dosing): 1 to 1.5 mg/kg (Ref).

Long surgical procedures (intermittent administration): Initial: 0.3 to 1.1 mg/kg; administer 0.04 to 0.07 mg/kg at appropriate intervals as needed.

Note: Pretreatment with atropine may reduce occurrence of bradycardia. Initial dose of succinylcholine must be increased when nondepolarizing agent pretreatment is used because of the antagonism between succinylcholine and nondepolarizing neuromuscular-blocking agents (Ref). When the cumulative dose of succinylcholine exceeds 2 to 4 mg/kg under general anesthesia, transition from a phase I to a phase II block may occur. If phase II block is suspected, diagnosis should be confirmed by peripheral nerve stimulation prior to administration of an anticholinesterase drug (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling. However, need for dosage adjustment unlikely as limited excretion in the urine.

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling.

Dosing: Obesity: Adult

The recommendations for dosing in patients with obesity are based upon the best available evidence and clinical expertise. Senior Editorial Team: Jeffrey F. Barletta, PharmD, FCCM; Manjunath P. Pai, PharmD, FCP; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC.

Class 1, 2 or 3 obesity ($\text{BMI} \geq 30 \text{ kg/m}^2$): **IV, IM:** No dosage adjustment necessary; use **actual body weight** for weight-based dose calculations (Ref). In patients with extreme obesity (eg, $\text{BMI} > 60 \text{ kg/m}^2$), consider using **adjusted body weight** for weight-based dosing calculations; however, intubating conditions may be less than optimal (Ref). Refer to adult dosing for indication-specific doses.

Rationale for recommendations:

Data evaluating succinylcholine dosing in extreme obesity originate from studies conducted in the surgery setting (Ref). In 1 study with healthy subjects, a positive correlation was noted between pseudocholinesterase activity and BSA, weight, and BMI. A similar duration of effect was observed for succinylcholine doses based on actual body weight, regardless of weight, after 1 mg/kg was administered (Ref). A retrospective cohort study suggests patients with obesity (≥ 120 kg) received a suboptimal dose (≤ 1 mg/kg) and subsequently had decreased first-pass success with rapid sequence intubation in the emergency department (Ref). A randomized, controlled trial evaluating succinylcholine weight-based dosing found use of actual body weight resulted in more favorable intubating conditions compared to using ideal body weight or lean body weight (Ref).

Dosing: Older Adult

Refer to adult dosing.

Dosing: Pediatric

(For additional information see "Succinylcholine (suxamethonium): Pediatric drug information")

Note: Dose to effect; doses will vary due to interpatient variability. To avoid prolonged neuromuscular blockade, use carefully and/or consider dose reduction in patients with reduced plasma cholinesterase activity due to genetic abnormalities of plasma cholinesterase or when associated with other conditions (eg, electrolyte abnormalities, neuromuscular disease, severe liver or kidney disease, malignant tumors). Initial dose of succinylcholine must be increased when nondepolarizing agent pretreatment used because of the antagonism between succinylcholine and nondepolarizing neuromuscular-blocking agents (Ref). Because of the risk of adverse effects including malignant hyperthermia and cardiac arrhythmias, surgical or long-term paralytic use (ie, continuous IV infusion) is not recommended (Ref).

Endotracheal intubation, emergent

Endotracheal intubation, emergent (eg, rapid sequence intubation): Note: To reduce the risk of bradycardia or asystole, premedication with atropine recommended prior to IV succinylcholine doses (Ref):

Note: Doses in obese patients should be calculated using total body weight (Ref).

IM:

Infants <6 months: Limited data available: 4 to 5 mg/kg; pharmacokinetic data suggests that young infants typically require doses on the higher end of this range (Ref).

Infants ≥ 6 months and Children: 4 mg/kg; maximum dose: 150 mg/dose (Ref).

Adolescents: 3 to 4 mg/kg; maximum dose: 150 mg/dose (Ref).

IV:

Infants ≤ 6 months: IV: 2 to 3 mg/kg/dose (Ref).

Infants >6 months: IV: 1 to 2 mg/kg/dose (Ref); doses up to 3 mg/kg/dose have been recommended (Ref).

Children: IV: 1 to 2 mg/kg/dose (Ref).

Adolescents: IV: 1 to 1.5 mg/kg/dose (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling.

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified.

Frequency not defined:

Cardiovascular: Abnormal T waves on ECG (peaked T-wave), asystole, bradycardia (higher with second dose; more frequent in children), cardiac arrhythmia, hypertension, hypotension, tachycardia, ventricular arrhythmia

Dermatologic: Skin rash

Endocrine & metabolic: Hyperkalemia

Gastrointestinal: Sialorrhea

Hypersensitivity: Anaphylaxis, hypersensitivity reactions

Nervous system: Malignant hyperthermia

Neuromuscular & skeletal: Fasciculations, jaw tightness, postoperative myalgia, rhabdomyolysis (can be acute; with possible myoglobinuric acute renal failure), state of neuromuscular blockade (with prolonged use: transition from phase I to phase II block)

Ophthalmic: Increased intraocular pressure

Respiratory: Apnea, respiratory depression

Miscellaneous: Tachyphylaxis

Postmarketing: Hypersensitivity: Nonimmune anaphylaxis

Contraindications

Hypersensitivity to succinylcholine or any component of the formulation; known or suspected genetic susceptibility to malignant hyperthermia; skeletal muscle myopathies; use after the acute phase of injury following major burns, multiple trauma, extensive denervation of skeletal muscle, or upper motor neuron injury.

Warnings/Precautions

Concerns related to adverse effects:

- Anaphylaxis: Severe anaphylactic reactions (some life-threatening and fatal) have been reported; immediate treatment (including epinephrine 1 mg/mL) for anaphylactoid and/or hypersensitivity reactions should be available during use. Use caution in patients with previous anaphylactic reactions to other neuromuscular blocking agents.
- Bradycardia: Risk of bradycardia may be increased with second dose and may occur more in children. Occurrence may be reduced by pretreating with anticholinergic agents (eg, atropine).
- Increased intraocular pressure: May increase intraocular pressure (IOP); avoid use in patients in which an increase in IOP is undesirable (eg, narrow-angle glaucoma, penetrating eye injuries) (Cunningham 1986; Gropper 2019).
- Intracranial pressure: May cause a transient increase in intracranial pressure (adequate anesthetic induction prior to administration of succinylcholine will minimize this effect).
- Intra gastric pressure: May increase intra gastric pressure, which could result in regurgitation and possible aspiration of stomach contents.
- Malignant hyperthermia: May trigger malignant hyperthermia; some reported cases have been fatal. Risk may be increased with concomitant administration of succinylcholine and volatile anesthetic agents and patients with genetic factors or family history of malignant hyperthermia, including ryanodine receptor or dihydropyridine receptor inherited variants. Signs of malignant hyperthermia may include arrhythmias, cyanosis, hemodynamic instability, hypercapnia, hyperthermia, hypovolemia, hypoxia, muscle rigidity, tachycardia, and tachypnea; coagulopathies, renal failure, and skin mottling

may also occur. If malignant hyperthermia is suspected, discontinue triggering agents and institute appropriate therapy (eg, dantrolene) and other supportive measures.

- Neuromuscular cross-sensitivity: Cross-sensitivity with other neuromuscular-blocking agents may occur; use extreme caution in patients with previous anaphylactic reactions to other neuromuscular-blocking agents.
- Vagal tone: May increase vagal tone.

Disease-related concerns:

- Burn injury: Use with caution in patients with extensive or severe burns; risk of hyperkalemia is increased following injury. Onset of time and duration of risk are variable, but risk is generally greatest 7 to 10 days after injury. Resistance may occur in burn patients ($\geq 20\%$ of total body surface area), usually several days after the injury, and may persist for several months after wound healing (Han 2009).
- Conditions that may *antagonize* neuromuscular blockade (decreased paralysis): Respiratory alkalosis, hypercalcemia, demyelinating lesions, peripheral neuropathies, denervation, muscle trauma, and diabetes mellitus may result in antagonism of neuromuscular blockade (ACCM/SCCM/ASHP [Murray 2002]; Greenberg 2013; Miller 2010; Naguib 2002).
- Conditions that may *potentiate* neuromuscular blockade (increased paralysis): Electrolyte abnormalities (eg, severe hypocalcemia, severe hypokalemia, hypermagnesemia), neuromuscular diseases, metabolic acidosis, respiratory acidosis, Eaton-Lambert syndrome, and myasthenia gravis may result in potentiation of neuromuscular blockade (Greenberg 2013; Miller 2010; Naguib 2002).
- Fractures/muscle spasm: Use with caution in patients with fractures or muscle spasm; initial muscle fasciculations may cause additional trauma.
- Hyperkalemia: Use with extreme caution in patients with preexisting hyperkalemia. Potentially severe hyperkalemia (increase up to ~ 0.5 mEq/L) may develop in patients with chronic abdominal infections, burn injuries, multiple trauma/crush injuries, extensive denervation of skeletal muscle, upper motor neuron injury, subarachnoid hemorrhage, or conditions which cause degeneration of the central and peripheral nervous system (Raman 1997; Roberts 2018; Weintraub 1969). However, in patients with a predisposition to hyperkalemia (eg, rhabdomyolysis, major trauma, burns) the increase can be >5 mEq/L, which may cause serious dysrhythmias (Gronert 2001; Roberts 2018).
- Plasma pseudocholinesterase disorders: Metabolized by plasma cholinesterase; use with caution (if at all) in patients suspected of being homozygous for the atypical plasma cholinesterase gene. Plasma cholinesterase activity may also be reduced by burns, anemia, decompensated heart disease, infections, malignant tumors, myxedema, pregnancy, severe hepatic or renal dysfunction, peptic ulcer, and certain medications and chemicals.

Special populations:

- Older adult: Use with caution in older adults, effects and duration are more variable.

Other warnings/precautions:

- Appropriate use: Maintenance of an adequate airway and respiratory support is critical. All patients should receive eye care including liberal use of lubricating drops, gel, or ointment and eyelids should remain closed during continuous neuromuscular blockade to protect against damage to the cornea (ulceration and dryness).
- Experienced personnel: Should be administered by adequately trained individuals familiar with its use.
- Risk of medication errors: Accidental administration may be fatal. Confirm proper selection of intended product, store vial so the cap and ferrule are intact and the possibility of selecting the wrong product is minimized, and ensure that the intended dose is clearly labeled and communicated, when applicable.

Warnings: Additional Pediatric Considerations

In children and adolescents, rare reports of acute rhabdomyolysis with hyperkalemia followed by ventricular dysrhythmias, cardiac arrest, and death have been reported in children with undiagnosed skeletal muscle myopathy, most frequently Duchenne muscular dystrophy. This syndrome presents as peaked T-waves and sudden cardiac arrest within minutes after the administration of succinylcholine. Bradycardia has been reported in both children and adults following succinylcholine and may progress to asystole; a higher incidence and severity of bradycardia occurs in pediatric patients. Bradycardia commonly occurs after initial doses >1.5 mg/kg in children; both children and adults are at risk following repeat doses. Also, if sudden cardiac arrest occurs immediately after administration of succinylcholine, consider hyperkalemia as potential etiology and manage accordingly. In neonates, some experts consider hyperkalemia a contraindication for use (AAP [Kumar 2010]).

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Injection, as chloride:

Anectine: 20 mg/mL (10 mL) [contains methylparaben (methyl p-hydroxybenzoate)]

Quelicin: 20 mg/mL (10 mL) [contains methylparaben (methyl p-hydroxybenzoate), propylparaben (propyl p-hydroxybenzoate)]

Generic: 20 mg/mL (10 mL)

Solution Prefilled Syringe, Injection, as chloride [preservative free]:

Generic: 100 mg/5 mL (5 mL); 200 mg/10 mL (10 mL)

Generic Equivalent Available: US

Yes

Pricing: US

Solution (Anectine Injection)

20 mg/mL (per mL): \$2.40

Solution (Quelicin Injection)

20 mg/mL (per mL): \$1.01

Solution (Succinylcholine Chloride Injection)

20 mg/mL (per mL): \$0.22 - \$2.40

Solution Prefilled Syringe (Succinylcholine Chloride Injection)

100 mg/5 mL (per mL): \$2.06

200 mg/10 mL (per mL): \$1.13

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Injection, as chloride:

Quelicin Chloride: 20 mg/mL (10 mL, 20 mL)

Generic: 20 mg/mL (10 mL, 20 mL)

Administration: Adult

IM: Administer deep IM only when IV access is not available.

IV: May be administered undiluted by rapid IV injection.

Prefilled syringe: For single use only. Expel air bubble and connect to appropriate injection connection; do not add any other fluid to syringe.

Administration: Pediatric

Parenteral:

IM: Injection should be made deeply. Use only when IV access is not available.

IV: May be administered undiluted by rapid IV injection.

Storage/Stability

Manufacturer recommends refrigeration of intact syringes and vials at 2°C to 8°C (36°F to 46°F) and may be stored at room temperature for 14 days; however, additional testing has demonstrated stability for ≤6 months unrefrigerated (25°C) (Ross 1988; Roy 2008). Stability in polypropylene syringes (20 mg/mL) at room temperature (25°C) is 45 days (Storms 2003). According to the 2020 to 2021 *ISMP Targeted Medication Safety Best Practices for Hospitals*, neuromuscular blockers should be segregated, sequestered, and differentiated from all other medication wherever they are stored (ISMP 2020).

Use: Labeled Indications

Neuromuscular blockade for endotracheal intubation, surgery, or mechanical ventilation:

As an adjunct to general anesthesia to facilitate tracheal intubation and to provide skeletal muscle relaxation during surgery or mechanical ventilation in adequately sedated patients.

Use: Off-Label: Adult

Electroconvulsive therapy, muscle relaxation

Medication Safety Issues

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication among its list of drug classes (neuromuscular blocker agent) which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Acute Care Settings ©ISMP 2024).

Other safety concerns:

According to the 2020 to 2021 *ISMP Targeted Medication Safety Best Practices for Hospitals*, neuromuscular blockers should be segregated, sequestered, and differentiated from all other medication wherever they are stored. This includes:

Only storing in places within the hospital that they are routinely used.

Placing in sealed boxes or in rapid sequence intubation kits (preferred).

Limiting availability in automated dispensing cabinets to perioperative, labor and delivery, critical care, and emergency departments only.

Placing in separate lidded containers within the pharmacy refrigerator or other isolated pharmacy storage area.

Affixing an auxiliary label to clearly communicate respiratory paralysis will occur and ventilation required on all storage bins and/or automated dispensing pockets/drawers (exception anesthesia-prepared syringes) stating one of the following:

Warning: Causes Respiratory Arrest - Patient Must Be Ventilated

Warning: Paralyzing Agent - Causes Respiratory Arrest

Warning: Causes Respiratory Paralysis - Patient Must Be Ventilated

International issues:

Quelicin [US, Brazil, Canada, Indonesia] may be confused with Keflin brand name for cefalotin [Argentina, Brazil, Mexico, Netherlands, Norway]

Metabolism/Transport Effects

None known.

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Acetylcholinesterase Inhibitors: May increase neuromuscular-blocking effects of Succinylcholine. *Risk C: Monitor*

Aminoglycosides: May increase therapeutic effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Aprotinin: May increase therapeutic effects of Succinylcholine. *Risk C: Monitor*

Bacitracin (Systemic): May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Bambuterol: May increase therapeutic effects of Succinylcholine. Bambuterol may increase serum concentrations of Succinylcholine. *Risk C: Monitor*

Beta-Blockers: May increase neuromuscular-blocking effects of Succinylcholine. *Risk C: Monitor*

Botulinum Toxin-Containing Products: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Bromperidol: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Capreomycin: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Cardiac Glycosides: Neuromuscular-Blocking Agents may increase arrhythmogenic effects of Cardiac Glycosides. *Risk C: Monitor*

Chloroquine: May increase neuromuscular-blocking effects of Succinylcholine. *Risk C: Monitor*

Clindamycin (Topical): May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Colistimethate: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. Management: If possible, avoid concomitant use of these products. Monitor for deeper, prolonged neuromuscular-blocking effects (respiratory paralysis) in patients receiving concomitant neuromuscular-blocking agents and colistimethate. *Risk D: Consider Therapy Modification*

Corticosteroids (Systemic): May increase neuromuscular-blocking effects of Succinylcholine. *Risk C: Monitor*

CycloPHOSphamide: May increase serum concentrations of Succinylcholine. Management: Consider alternatives to succinylcholine in patients who have received cyclophosphamide in the past 10 days. If combined, consider reduced succinylcholine doses and close monitoring for increased and prolonged neuromuscular blocking effects. *Risk D: Consider Therapy Modification*

CycloSPORINE (Systemic): May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Desflurane: May increase neuromuscular-blocking effects of Succinylcholine. *Risk C: Monitor*

Echothiophate Iodide: May increase serum concentrations of Succinylcholine. Management: For patients receiving echothiophate iodide eye drops, consider using a neuromuscular-blocking agents other than succinylcholine. If succinylcholine is used, consider a reduced dose, and monitor for enhanced/prolonged effects. *Risk D: Consider Therapy Modification*

Esketamine (Injection): May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Estrogen Derivatives: May increase serum concentrations of Succinylcholine. *Risk C: Monitor*

Gepotidacin: May increase therapeutic effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Isoflurane: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Ketamine: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Lincosamide Antibiotics: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Lithium: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Local Anesthetics: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Loop Diuretics: May decrease neuromuscular-blocking effects of Neuromuscular-Blocking Agents. Loop Diuretics may increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Magnesium Salts: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Metoclopramide: May increase neuromuscular-blocking effects of Succinylcholine. *Risk C: Monitor*

Minocycline (Systemic): May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Opioid Agonists: Succinylcholine may increase bradycardic effects of Opioid Agonists. *Risk C: Monitor*

Oxytocin: May increase neuromuscular-blocking effects of Succinylcholine. *Risk C: Monitor*

Pancuronium: May increase adverse/toxic effects of Neuromuscular-Blocking Agents. Pancuronium may increase therapeutic effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Phenelzine: May increase neuromuscular-blocking effects of Succinylcholine. Management: Consider a lower initial succinylcholine dose in patients taking phenelzine, or those who have taken phenelzine within the previous 2 weeks. Monitor for increased succinylcholine effects (eg, prolonged apnea) in patients who recently received phenelzine. *Risk D: Consider Therapy Modification*

Pholcodine: May increase adverse/toxic effects of Neuromuscular-Blocking Agents. Specifically, anaphylaxis has been reported. *Risk C: Monitor*

Polymyxin B: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Procainamide: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Promazine: May increase neuromuscular-blocking effects of Succinylcholine. *Risk C: Monitor*

QuiNIDine: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

QuiNINE: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk X: Avoid*

Sertraline: May increase serum concentrations of Succinylcholine. *Risk C: Monitor*

Terbutaline: May increase neuromuscular-blocking effects of Succinylcholine. *Risk C: Monitor*

Tetracyclines: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Vancomycin: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Reproductive Considerations

Sensitivity to succinylcholine may be increased due to a decrease in plasma cholinesterase activity in patients taking chronic oral contraceptives.

Pregnancy Considerations

Succinylcholine crosses the placenta.

An increased risk of major congenital anomalies has not been observed. Sensitivity to succinylcholine may be increased due to a ~24% decrease in plasma cholinesterase activity during pregnancy and several days postpartum; prolonged maternal apnea may be observed. Newborns of mothers with atypical plasma cholinesterase or those exposed to repeated or high doses of succinylcholine during cesarean delivery should be monitored for apnea and flaccidity.

Breastfeeding Considerations

It is not known if succinylcholine is present in breast milk.

According to the manufacturer, the decision to breastfeed during therapy should consider the risk of infant exposure, the benefits of breastfeeding to the infant, and the benefits of treatment to the mother.

Monitoring Parameters

Vital signs (heart rate, BP, respiratory rate); oxygen saturation and end-tidal carbon dioxide (if available) (Roberts 2018); degree of muscle paralysis (eg, presence of spontaneous movement, ventilator asynchrony, shivering, and consider use of a peripheral nerve stimulator with train of four monitoring along with clinical assessments); serum potassium and calcium.

Mechanism of Action

Acts similar to acetylcholine, produces depolarization of the motor endplate at the myoneural junction which causes sustained flaccid skeletal muscle paralysis produced by state of accommodation that develops in adjacent excitable muscle membranes

Pharmacokinetics (Adult Data Unless Noted)

Onset of action: Dependent on route, age, and dose; data suggest faster onset with higher doses (Coté 2019):

IM: Infants and Children: 3 to 4 minutes (Liu 1981); Adults: 2 to 3 minutes.

IV:

Neonates and Infants: ~30 seconds (range: 19 to 40 seconds [dose: 2 to 4 mg/kg]) (Meakin 1990).

Children and Adolescents: 35 to 55 seconds (Coté 2019).

Adults: Flaccid paralysis: <60 seconds.

Duration: Dependent on route, age, and dose; hypothermia may prolong the duration of action.

IM: 10 to 30 minutes; Observed to be shorter in infants than children.

IV: ~4 to 10 minutes; Faster recovery rate in infants and children compared to adults (Fisher 1975; Greenberg 2013).

Distribution: V_d higher in neonates and infants due to larger ECF volume; higher IV doses necessary.

Metabolism: Rapidly hydrolyzed by plasma pseudocholinesterase to inactive metabolites.

Excretion: Urine (~10% excreted unchanged).

Patient Counseling Points

(For additional information see "Succinylcholine (suxamethonium): Patient drug information")

What is this drug used for?

- It is used to relax muscles while on a breathing machine or during surgery.

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

- High potassium levels like a heartbeat that does not feel normal; feeling confused; feeling weak, lightheaded, or dizzy; feeling like passing out; numbness or tingling; or shortness of breath
- High or low blood pressure like very bad headache or dizziness, passing out, or change in eyesight
- Chest pain or pressure
- Fast, slow, or abnormal heartbeat
- Muscle pain or weakness, dark urine, or trouble passing urine
- Twitching
- Change in eyesight or eye pain
- Drooling
- Malignant hyperthermia like a fast or abnormal heartbeat; fast breathing; fever; spasm or stiffness of the jaw muscles; blue or gray skin; patchy skin; unexplained bruising or bleeding; or kidney problems like unable to pass urine, change in how much urine is passed, blood in the urine, or a big weight gain
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from

the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Midarine;
- (AR) Argentina: Succinilcolina fabra | Succinilcolina gray;
- (AT) Austria: Lysthenon;
- (BD) Bangladesh: Rapilax;
- (BE) Belgium: Celocurine | Myoplegine;
- (BR) Brazil: Quelicin | Succitrat;
- (CN) China: Suxamethonium;
- (CO) Colombia: Quelicin;
- (DE) Germany: Lysthenon | Pantolax | Succicuran;
- (EC) Ecuador: Quelicin;
- (EE) Estonia: Sukolin;
- (EG) Egypt: Succinyl choline;
- (ES) Spain: Anectine | Mioflex;
- (FI) Finland: Sukolin | Suxamethonium chloride panpharma | Suxamethonium ethypharm;
- (FR) France: Biocodex suxamethonium | Suxamethonium aguettant | Suxamethonium ethypharm | Suxamethonium panpharma;
- (GB) United Kingdom: Anectine | Scoline;
- (HU) Hungary: Sukolin;
- (IE) Ireland: Anectine | Suxamethonium;
- (IN) India: Suxim;
- (KR) Korea, Republic of: Succicholine;
- (KW) Kuwait: Midarine;
- (LB) Lebanon: Suxamethonium;
- (LU) Luxembourg: Myoplegine;
- (MX) Mexico: Anectine;
- (NL) Netherlands: Curalest | Suxamethoniumchloride CF;
- (NO) Norway: Lysthenon | Suxamethonium | Suxamethonium chloride dihydrate ethypharm | Suxamethonium chloride martindale;
- (NZ) New Zealand: Ethicholine | Succinolin;
- (PH) Philippines: Anectine;
- (PK) Pakistan: Succinyl | Suxam;
- (PL) Poland: Anectine | Lysthenon | Succinylcholinjodid;
- (PR) Puerto Rico: Anectine | Quelicin | Quelicin-1000 | Succinyl choline;
- (PT) Portugal: Suxametonio;
- (QA) Qatar: Anectine | Lysthenon | Lysthenon Forte | Midarine | Succinolin;
- (RU) Russian Federation: Suksametonij biolek;
- (SE) Sweden: Celocurin;
- (SI) Slovenia: Succinylcholin;
- (TR) Turkey: Lysthenon;
- (TW) Taiwan: Scoline | Succinyl;
- (ZA) South Africa: Micro-suxamethonium | Pharma Q Suxamethonium | Q-med suxamethonium | Suxamethonium

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